Bi3D++: Hybrid Bi-Domain Active Learning for Cross-Domain 3D Object Detection

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***Abstract*—Domain adaptation has recently been widely explored for 3D detection. Previous works mainly use unsupervised domain adaptation (UDA) to address domain discrepancies. Despite notable improvements, their performance still largely trails models trained with fully annotated target data, due to larger domain gaps caused by different sensors and changing environments. In this paper, we exploit key characteristics of autonomous driving scenarios, including similar scenes and classimbalanced distributions, and explore a new task named active domain adaptation (ADA) for 3D object detection, which selects partial but important target data for annotation to further improve target-domain performance. Such a setting better reflects practical deployment in practice, where annotating all target-domain point clouds is prohibitively expensive while limited labels can substantially guide adaptation effectively. To this end, we propose a hybrid bi-domain active learning strategy, Bi3D++, to sample valuable data from both source and target domains and transfer source-domain knowledge to the target domain. Bi3D++ first samples target-like source data by measuring scene-level and instance-level similarity between domains, avoiding interference from irrelevant source data. Then, a hybrid active target sampling strategy selects target data by jointly considering rare-class similarity, intra-frame diversity, and inter-frame diversity, enabling diverse frames with diverse instances while emphasizing rare classes. Experiments on multiple cross-domain settings, including cross-beam and cross-location, show that Bi3D++ outperforms state-of-theart UDA methods with only 1% labeled target data and consistently improves performance as target annotations increase.**

***Index Terms*—3D Object Detection, cross-domain, active learning, point clouds, Bi-domain, autonomous driving.**

I. Introduction

To DATE, LiDAR-based 3D object [1], [2], [3], [4] detection has attracted extensive attention due to its practical applications, such as autonomous driving (AD) and robotics. However, these state-of-the-art 3D detection methods perform well on the single domain or the source domain (*e.g.*, Waymo dataset [5]) and hard to generalize to the new target domain (*e.g.*, KITTI dataset [6]) due to domain shifts such as the differences in sensors and changes of environments. Such a problem is defined as cross-domain 3D object detection, which has been explored in recent years [7], [8], [9], [10]. In real-world applications, the continuous iterative upgrade of sensors and dynamic changes in the surrounding environments make cross-domain 3D object detection an important issue and an urgent problem to be solved.

Recently, addressing domain discrepancies in 3D object detection has garnered significant attention, with existing works broadly categorized into two types. The first type draws inspiration from the unsupervised domain adaptation (UDA) technique used in 2D cross-domain tasks [11], [12], [13], [14], [15], [16], [17], and attempts to transfer UDA methods such as self-training to autonomous driving scenarios [18], [19], [20], [21], [22]. Among them, ST3D family [7], [23] introduces a self-training pipeline that continuously updates pseudo labels (i.e., using filtered detector predictions as training labels) and finally adapts a pre-trained detector from the source domain to a new target domain. ReDB [10] proposes a self-training-based pipeline to alleviate cross-domain 3D object detection problems under multi-class settings. To mitigate the challenges posed by inaccurate pseudo labels in UDA methods, another line of research leverages some labeled target domain data to reduce noisy supervision and further boost performance on the target domain. For instance, SSDA3D [24] is the first to explore the semi-supervised domain adaptation (SSDA) tasks in 3D object detection, achieving substantial improvements in detection accuracy.

Despite notable performance improvements, these DA models still face the following challenges. 1) The performance of existing methods still largely falls behind that of supervised methods (i.e., trained on a fully labeled target domain). This is primarily because UDA methods are hindered by noisy pseudo-labels, while SSDA methods are prone to sampling redundant labels, which can severely impact the performance, especially when the number of labeled data is relatively small. 2) Most existing methods [7], [8], [9], [21] except ReDB [10] focus on a single class (i.e., car), which is the most common category in AD scenarios. This is because transferring knowledge for such a category is relatively easier. However, in practical applications, other objects (*e.g.*, pedestrians, cyclists) in road scenes are equally critical to ensure safety.

To alleviate these problems, inspired by active learning [25], we consider selecting partial yet important target data to

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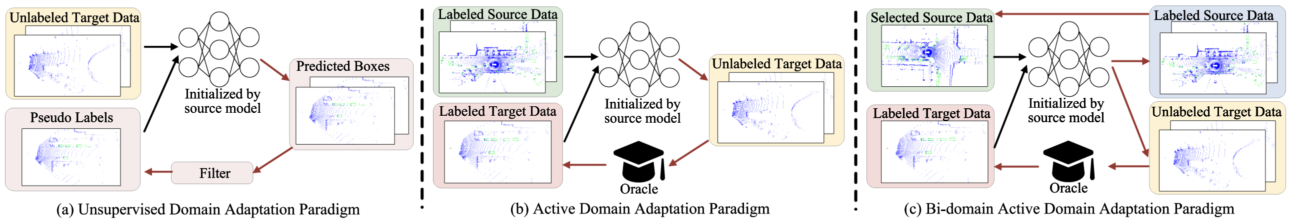


Fig. 1. Different paradigms of (a) unsupervised domain adaptation, (b) active domain adaptation, and (c) bi-domain active domain adaptation. Previous active domain adaptation works only focus on selecting target domain data. In contrast, our bi-domain active learning methods select both source and target data. The black arrow and red arrow denote the training flow and the active sampling flow, respectively.

TABLE I
THE AVERAGE NUMBER OF EACH CLASS PER FRAME

Dataset	Car / Vehicle	Pedestrian	Cyclist
KITTI [6]	3.8	0.60	0.21
nuScenes [58]	12.08	5.75	0.291
Waymo [5]	30.22	14.10	0.33

TABLE II
THE PERFORMANCE (AP_{BEV}/AP_{3D}) OF RANDOM SAMPLING 1% TARGET DATA ON WAYMO → KITTI (W → K) AND NUSCENES → KITTI (N → K) SCENARIOS. DIRECTLY MEANS THE SOURCE ONLY RESULT AND RANDOM MEANS THE RESULTS AFTER TRAINING ON RANDOM SAMPLING DATA. NOTE THAT THE RANDOM SAMPLING RESULTS ARE OBTAINED BY AVERAGING OVER THREE EXPERIMENTAL RUNS.

Setting	Car	Pedestrian	Cyclist
W → K (Directly)	58.25 / 10.48	51.58 / 49.06	51.97 / 49.10
W → K (Random)	80.42 / 70.04	47.98 / 41.92	40.65 / 35.89
N → K (Directly)	59.49 / 25.67	17.77 / 14.01	11.55 / 9.61
N → K (Random)	74.41 / 62.12	15.71 / 13.22	4.55 / 3.78

perform manual annotation and train the model on selected data. This approach not only ensures the existence of correct supervision but also avoids redundancy in labeled data. Such a paradigm is called active domain adaptation (ADA), as shown in Fig. 1(b), has already been explored in 2D classification tasks (*e.g.*, TQS [26] and CLUE [27]). However, in the field of 3D vision, this task remains under-explored. By directly applying widely-used 2D ADA methods such as TQS [26] and CLUE [27] to several typical 3D detectors (*e.g.*, PV-RCNN [28] and SECOND [29]), we observed that these 2D ADA methods cannot achieve satisfactory performance in 3D scenarios. As shown in Table III and Table IV, for example, PV-RCNN coupled with TQS only achieves 66.81% / 61.39% AP_{BEV} / AP_{3D}, which largely falls behind the fully-supervised result (i.e., 72.25% / 67.71%).

The main obstacle to using 2D ADA methods to tackle domain discrepancies in 3D scenes can be attributed to the following three reasons: 1) *Larger domain discrepancies in 3D scenario.* Unlike 2D images, LiDAR point clouds can be severely affected by the parameters of LiDAR sensor (*e.g.*, beams), making it difficult for the source model to learn the domain-invariant knowledge and adapt to the target domain. Further, the widespread intra-domain feature variations in the source domain widen the gap between the selected target domain samples and the entire source domain samples, leading to negative transfer to the target domain. 2) *The similar instances and scenes in autonomous driving scenarios.* The adjacent frames and instances with similar distance and relative position to the LiDAR are similar, which may lead to sampling similar frames or frames including similar instances using 2D ADA methods, resulting in overfitting to similar frames or instances and redundant annotations. 3) *Rare-class problem.* The distribution of categories in road scenes presents a long-tail distribution and previous 2D ADA methods tend to ignore the tail categories while sampling. As a result, the model will overfit to the dominant class and partially ignore other classes, which are also important in driving scenes.

To this end, we propose Bi3D++, a hybrid bi-domain active learning framework for 3D object detection tasks as shown in Fig. 1(c). Bi-domain means that, unlike previous ADA methods that focus on selecting data from a single target domain, Bi3D++ samples valuable data from both the source and the target domain. First, to alleviate the larger domain discrepancies in 3D scenarios, we design a hybrid active source sampling strategy that aims to select target-like samples by comprehensively considering both instance-level and scene-level similarity. The source model is then trained on the selected source samples, which have smaller domain gaps with the target domain distribution and can thus be efficiently transferred to the target domain without accessing the target labels. Then, to address the issues of redundant sampling and class imbalance, we design a diversity-based hybrid active target sampling strategy which is composed of three criteria (i.e., rare-class-aware sampling criterion, intra-frame diversity criterion, and inter-frame diversity criterion). By measuring the instance similarity of target RoI features and rare-class prototypes, the rare-class similarity criterion enables the model to identify frames containing more tail classes. Additionally, by analyzing both the distribution of the RoI features within the same class and the similarity across frames, the intra-frame and inter-frame diversity criteria address two key challenges: overfitting and redundant annotation caused by sampling similar instances and frames. Finally, by training on both selected source and target data, the detector to further adapt to the new target domain at a low cost.

Compared to our previous CVPR conference version [30], this version (termed Bi3D++) has improved in the following aspects: 1) We extend our method to multiple classes (i.e., car, pedestrian, cyclist). 2) We design a hybrid source sampling strategy that effectively selects target-like source data by comprehensively measuring both scene-level domainness and instance-level similarity. This approach helps in filtering out irrelevant source data and adapting the detector to the target

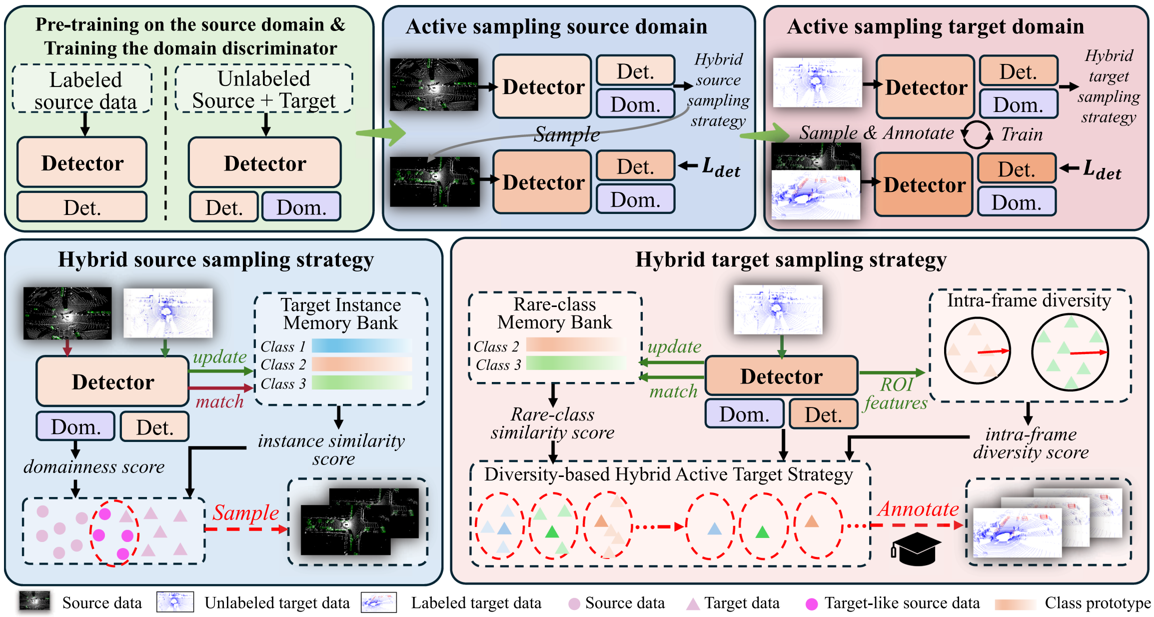


Fig. 2. Overview of Bi3D++. The upper section displays the overall workflow of Bi3D++, and the lower section elaborates on the two core hybrid sampling strategies. In the active sampling source stage, labeled source data and unlabeled target data are first fed into the detector and the target-like source data are selected based on scene-level domainness and instance-level similarity. In the active sampling target stage, Bi3D++ selects target data based on the designed rare-class-aware similarity and diversity criteria. Then, an oracle (usually a human annotator) annotates the selected target data, which, alongside selected source data, trains the detector to improve target domain performance. Note that G_i represents the group i clustered based on the scene-level features which will be detailed in Section III-C.

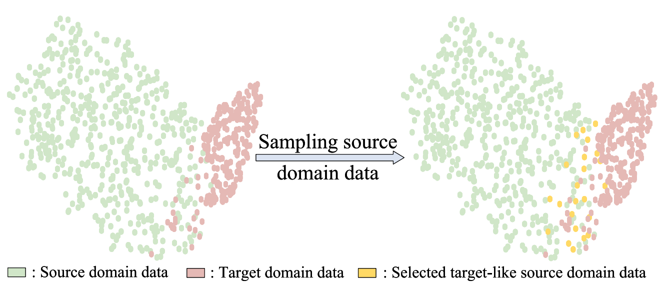


Fig. 3. The visualization of the distribution of the source and target domains. We sample target-like source domain data using the hybrid source sampling strategy.

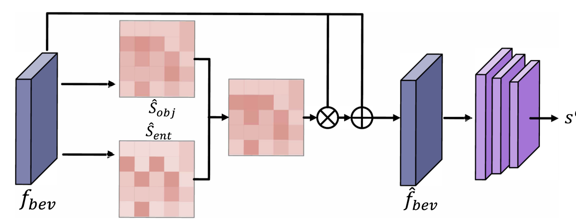


Fig. 4. Foreground-aware scene-level domain discriminator.

domain. 3) We propose a hybrid target sampling strategy that integrates a rare-class-aware sampling criterion with intra-frame and inter-frame diversity criteria. This strategy is designed to sample diverse frames containing diverse instances while specifically addressing the class-imbalance distribution (e.g., rare classes like pedestrians and cyclists) in autonomous driving scenarios. 3) We provide a detailed analysis of combining different cross-domain methods (i.e., UDA, and SSDA) and conduct experiments on label-insufficient to label-rich settings and settings with different detector architectures. 4) We boost the performance compared to the conference version and experimental results show that the proposed methods surpass all UDA methods using only 1% target samples on all classes.

The remainder of this paper is organized as follows. In Section II, we introduce related works of Bi3D++. In Section III, we detail the proposed method. In Section IV, we provide extensive experiments and analysis. In Section V, we conclude this paper.

II. Related Works

Our work is closely related to LiDAR-based 3D object detection, active domain adaptation, and domain adaptation for 3D object detection. We review the related works in the three fields and compare our method with the existing methods in the following part.

A. LiDAR-Based 3D Object Detection

With the development of autonomous driving and robotics, LiDAR-based 3D object detection has attracted widespread attention. Different types of works have been proposed to achieve better accuracy. These works can be roughly divided into three categories: voxel-based methods, point-based methods, and point-voxel-based methods. Voxel-based methods [2], [29], [31], [32], [33] first divide irregular point clouds into ordered grids and then extract features using 3D and 2D backbones. SECOND [29] is a pioneer of voxel-based methods that explores sparse convolution networks in 3D object detection and greatly improves the efficiency of the detector. PointPillars [1] designs a pillar-based method that only uses 2D convolution to extract features and largely improves the efficiency of the detector. More recently, SST [34] and DSVT [35] explored 3D detection with transformer architectures by designing different attention mechanisms. In contrast, point-based methods [36] directly extract features and generate bounding boxes from raw points. PointRCNN [36] is a prior work that generates 3D RoIs with foreground segmentation and designs an RCNN-style two-stage refinement. IA-SSD [37] proposes a single-stage point-based detector that reduces memory and computational cost. To combine the advantages of voxel-based methods and point-based methods, point-voxel-based methods [28], [38] (i.e., PV-RCNN, PV-RCNN++) propose a point-voxel set abstraction to fuse point and voxel features and greatly improve the accuracy on multiple datasets. However, these state-of-the-art 3D detectors are unstable when encountering domain discrepancies, resulting in performance drops when applied in dynamically changing environments.

B. Active Domain Adaptation

Active Learning assumes that different samples have different values and tries to select a small number of samples with the highest values to construct a training set to achieve high accuracy with few labeled samples. The value quantifies the benefit of annotating each unlabeled data instance and most active learning methods estimate the value using the designed acquisition function, such as committee [39], uncertainty [40], and diversity [41]. Committee-based methods [39] use multiple classifiers as a committee and select samples on which the classifiers disagree the most. Uncertainty-based methods [40] select samples with the most uncertainty, which are usually evaluated by entropy. Diversity-based methods [41] expect the distribution of selected samples to be consistent with the whole dataset. To sample more valuable data, recent works [42], [43], [44] often combine multiple criteria to take advantage of different approaches. For example, Wu et al. [42] design an active learning framework for 2D object detection by combining uncertainty-based methods (i.e., using entropy to evaluate uncertainty) and diversity-based methods. BiLAF [44] adopts a bi-level framework that not only maintains global diversity via core sample selection but also captures local uncertainty by identifying samples near decision boundaries. TAILSMAN [45] utilizes submodular mutual information to handle the long-tail distribution during active sampling. More recently, researchers began to explore active learning in 3D scenarios. As a pioneer, CRB [46] explores active learning in 3D object detection. It comprehensively considers label conciseness, feature representativeness, and geometric balance. KECOR proposes an information-theoretic-based criterion that can achieve comparable performance with fewer sampling boxes.

Active Domain Adaptation (ADA) [26], [27], [47], [48], [49] aims to transfer knowledge from a source domain to a target domain by selecting the most informative target samples for annotation. Existing ADA methods generally follow the principles of active learning methods while explicitly considering domain discrepancies to avoid selecting outliers or negative transfer samples. CLUE [27] draws inspiration from diversity and uncertainty-based methods. It argues that direct uncertainty sampling is sensitive to domain shifts and thus proposes an uncertainty-aware clustering algorithm to ensure both representativeness and informativeness. TQS [26] incorporates committee and uncertainty mechanisms but specifically addresses domain discrepancies. It designs a mixed sampling criterion that combines committee disagreement, uncertainty, and domainness to filter out samples that are hard to adapt. Furthermore, SDM [47], EADA [48], and DUC [50] align with the uncertainty-based paradigm but innovate by mining the intrinsic properties of the model or calibrating the uncertainty estimation. Specifically, SDM utilizes the maximum margin loss, and EADA leverages free energy. DUC employs a Dirichlet-based approach to treat predictions as distributions, ensuring more reliable uncertainty estimation for sample selection.

However, most existing works explore active learning and active domain adaptation in 2D scenarios, which are not suitable for 3D scenarios. Besides, the domain discrepancies are different in 2D and 3D scenarios, which further makes it impossible to directly apply 2D methods to 3D scenarios.

C. Domain Adaptation for 3D Object Detection

Thanks to the rapid development of 3D vision and domain adaptation on object detection [51], [52], domain shifts in 3D scenarios are attracting increasing attention. Several works [7], [8], [9], [10], [23], [53], [54], [55], [56], [57] have begun to explore the domain-adaptable or few-shot in 3D object detection. As a pioneer, SN [53] analyzes the domain gaps in autonomous driving scenarios and finds that the distribution of object size severely reduces the performance of the detectors in the target domain. ST3D [7] and ST3D++ [23] explore self-training methods in domain adaptive 3D object detection and propose a self-training pipeline that generates pseudo labels of the target domain to mitigate the domain gaps. LiDAR distillation [9] combines distillation and LiDAR beam downsampling, efficiently transferring knowledge from high-beam data to low-beam data. GPA-3D [8] aligns BEV features from the source and target domains to reduce domain gaps. SSDA3D [24] first explores SSDA for 3D object detection and proposes a two-stage domain adaptation method consisting of an inter-domain adaptation stage and an intra-domain generalization stage. However, the above methods only focus on a single class (i.e., car) and ignore other classes, such as pedestrian and cyclist, which are also important in road scenes. More recently, ReDB [10] revisits previous self-training methods and proposes a balanced pseudo method that can greatly improve the performance in multi-category scenarios.

Although these UDA methods boost the performance on the target domain, the performance gaps between UDA methods and fully supervised ones are still large and meanwhile, most existing methods are designed for a single class, which ignores other rare classes that are also essential in road scenes. However, our Bi3D++ samples a small amount of target domain data to annotate and largely improves the performance of three common classes at the same time. Besides, Bi3D++ is orthogonal to UDA and SSDA methods, which will be further analyzed in Section IV-C2.

III. Bi3D++

In this section, we introduce the design of Bi3D++. Our Bi3D++ tackles the domain adaptive 3D object detection tasks by active sampling both source and target data and training the detector on sampled data. The overall framework of Bi3D++ is shown in Fig. 2. To better illustrate our proposed method, in the following part of this section, we first provide the problem definition and baseline introduction in Section III-A. After that, we detail the source sampling strategy and target sampling strategy in Section III-B and Section III-C respectively. Finally, we elaborate on the Bi3D++ training objectives and training and sampling process in Section III-E.

A. Problem Formulation

Given a labeled source domain dataset $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ and an unlabeled target domain dataset $D_t = \{x_i^t\}_{i=1}^{n_t}$ where n_s and n_t represent the number of source and target data, respectively. Our goal is to select a small amount of target data and obtain the labels by the oracle (usually a human annotator) to adapt the model trained on D_s to D_t by training on the selected labeled target data. The selected data are sampled in N rounds to construct the labeled target set $D_t' = \{(x_i^t, y_i^t)\}_{i=1}^{B}$ where B is the annotation budget and $B \ll n_t$. In the k-th active sampling round where $k \in [1, N]$, b_k samples are selected from $D_t / D'_{t,k-1}$ where $D'_{t,k-1}$ denotes the selected data of previous $k-1$ rounds and $D'_{t,0} = \emptyset$. The $D'_{t,k}$ can be obtained by combining $D'_{t,k-1}$ and b_k selected samples of the current round. Then, the model is trained on $D'_{t,k}$. Finally, the labeled target set is acquired after N rounds of sampling and the number of labeled data reaches B (i.e., $\sum_{i=1}^{N} b_k = B$).

Different from the previous ADA methods which focus on how to select target data, Bi3D++ also takes selecting source data into consideration. This is because the source domain data are widely distributed and only part of them are beneficial for transferring the knowledge learned from the source domain to the target domain. More specifically, a selected source set D_s' is constructed during the active source sampling process and the model will be trained on the selected source data to improve the performance on the target domain.

B. Hybrid Active Source Sampling Strategy

Bi3D++ first utilizes a hybrid active source sampling strategy to select target-like source samples to adapt the detector to the target domain before accessing the selected labeled target data. As shown in Fig. 3, a distribution gap exists between the source and target domains due to domain discrepancies. However, certain source data are distributed closer to the target domain, which motivates us to select these target-like source data to improve the detector's performance on the target domain. Furthermore, the discrepancies between different domains can be attributed to two factors: scene-level differences caused by variation in collection locations and equipment and instance-level differences caused by the inconsistent statistical distribution of objects. To this end, we propose a hybrid active source sampling strategy that incorporates scene-level and instance-level similarity between the source and target domains to effectively sample the target-like source domain data.

1) Foreground-Aware Scene-Level Domainness: To effectively evaluate the similarity between scenes, we first need to extract scene representations. However, due to the sparsity of point clouds, it is essential to focus more on the foreground regions to avoid interference from a large amount of background regions, thereby obtaining effective scene-level representations. Therefore, we design a foreground-aware scene-level domain discriminator as shown in Fig. 4 to measure the domainness of each frame based on the Bird-Eye-View (BEV) features. In detail, given input point cloud $\mathbf{x}_i^d$ where $d \in [s, t]$ denotes the source and the target domain, respectively. The raw point clouds are first encoded by 3D and 2D backbones and then mapped to BEV space to obtain the BEV features $f_{bev}^d \in \mathbb{R}^{C\times H\times W}$, where C is the channel number, H and W are the height and width of the BEV feature, respectively.

To make the discriminator focus on the foreground regions, we first calculate the objectness score $S_{obj} \in \mathbb{R}^{C'\times H\times W}$ of each spatial location of the current BEV feature f_{bev} using RPN where C' denotes the number of anchors at each spatial location. A higher objectness score indicates that there is a higher probability that the current position is a foreground region. However, relying solely on the objectness score cannot extract accurate foreground information of both the source and target domains due to the domain discrepancies (*e.g.*, the model may be unable to reliably determine whether the current target region is a foreground region). To address this issue, we further calculate the entropy score $S_{ent} \in \mathbb{R}^{C'\times H\times W}$ which measures the model's uncertainty of the current region using the following formula:

$$S_{ent} = -S_{obj}\log S_{obj} - (1 - S_{obj})\log(1 - S_{obj}). \quad (1)$$

Based on (1) and previously obtained S_{obj}, we generate a scene-level attention map by combining S_{obj} and S_{ent}, which can make the model pay more attention to foreground regions. Thus, the foreground region-aware features can be calculated as follows:

$$\hat{f}_{bev}^d = \left(1 + \frac{\hat{S}_{obj} + \hat{S}_{ent}}{2}\right) f_{bev}^d, \quad (2)$$

here we use residual connection to avoid losing information and $\hat{f}_{bev}^d$ denotes the foreground region-aware BEV features, and $\hat{S}_{obj}$ and $\hat{S}_{ent}$ represent the maximum value of S_{obj} and S_{ent} along the channel dimension, respectively.

After obtaining $\hat{f}_{bev}^d$, a CNN-based domain discriminator is used to classify whether the current frame belongs to the source domain or the target domain. We use a binary cross-entropy loss to optimize the domain discriminator, which can be written as follows:

$$\begin{aligned} L_{dis} = & -\mathbb{E}_{x^s\sim D_s}\left[\log(1 - H(\hat{f}_{bev}^s)\right] \\ & -\mathbb{E}_{x^t\sim D_t}\left[\log(H(\hat{f}_{bev}^t)\right], \end{aligned} \quad (3)$$

where L_{dis} is the loss of domain discriminator H, and s/t represents source or target domain. In the training process, we label the source domain and the target domain as '0' and '1' to calculate L_{dis}, respectively. Furthermore, the domainness score

$s^d \in [0, 1]$ can be given by the domain discriminator, and a domainness score close to 1 represents that the current frame is closer to the distribution of the target domain and vice versa.

2) Memory-Based Instance-Level Similarity Assessment: However, only calculating scene-level domainness may overlook the instance features that are also essential in object detection tasks. Therefore, we propose a memory-based instance-level similarity assessment approach to further consider the instance-level feature similarity when selecting target-like source data. Specifically, we first construct a target instance memory bank $B_t = \{p^c\}_{c=1}^{N_c}$ which stores the prototypes of each class in the target domain and is initialized as empty where p^c denotes the prototype of class c and N_c is the number of classes. Given an unlabeled target data x_i^t, the RoI features $\mathbf{r}_i^t = \{r_{i,j}^t\}_{j=1}^{N_{RoI}}$ and its corresponding category $\mathbf{c}_i^t = \{c_{i,j}^t\}_{j=1}^{N_{RoI}}$ can be predicted by the detector where $r_{i,j}^t$ and $c_{i,j}^t$ are the feature and predicted class of the j-th RoI and N_{RoI} is the number of RoI in the current frame. Then, we can obtain the prototype p_i^c of each class (i.e., a single prototype that represents the category for the specific frame) in the current frame by averaging RoI features per category. The prototypes in B_t can be updated as follows:

$$p^c \leftarrow \alpha p^c + (1 - \alpha) p_i^c, \tag{4}$$

where α is a hyperparameter which determines the updating step.

After B_t is acquired, the similarity between source RoIs and target RoI prototypes can be calculated by cosine similarity. In detail, given source RoI features $\mathbf{r}_i^s = \{r_{i,j}^s\}_{j=1}^{N_{RoI}}$ and its corresponding category $\mathbf{c}_i^s = \{c_{i,j}^s\}_{j=1}^{N_{RoI}}$, the similarity can be calculated by:

$$\mathbf{Sim}_i^c = \frac{p^c \cdot \mathbf{r}_i^s}{\|p^c\| \|\mathbf{r}_i^s\|}, \;\; p^c \in B_t, \tag{5}$$

where $\mathbf{Sim}_i^c \in \mathbb{R}^{1 \times N_{RoI}}$ represents the similarity between the prototype of class c in the target domain and current source RoIs. Finally, the instance similarity can be calculated by the average of the max similarity value of each class which can be written as follows:

$$s_i^{sim} = \mathbb{E}_{c \in [1, N_c]} \max(\mathbf{Sim}_i^c), \tag{6}$$

where $s_i^{sim} \in [0, 1]$ denotes the similarity at instance-level.

3) Hybrid Active Source Strategy: The goal of active source sampling is to select target-like source data and improve model performance on the target domain by training on the sampled source data. To better select target-like source data, we consider similarity at both scene and instance levels. Specifically, the domainness score s_i^d and the instance similarity s_i^{sim} can be given by the domain discriminator and (6). The final score of i-th source frame can be obtained by adding the two scores (i.e., $s_i^{final} = s_i^d + s_i^{sim}$). A higher score means that the current frame is closer to the distribution of the target domain in both scene-level and instance-level. Then, by sorting the source data according to the final score in descending order, we can sample the top-p_{src} percent source data to construct D_s'. The algorithm of hybrid active source sampling is summarized in Algorithm 1.

Algorithm 1: Hybrid Active Source Strategy.

Input: Source dataset D_s and unlabeled target dataset D_t.
Output: The selected source set D_s'.

```
Train the domain discriminator using Eq. (3).
Initialize the target instance memory bank B_t by empty.
for x_i^t in D_t do
    Calculate the prototype of each class and update the B_t using the Eq. (4).
for x_i^s in D_s do
    Calculate the domainness score s_i^d and similarity score s_i^sim using domain discriminator and Eq. (5).
    Calculate the final score s_i^final of the current frame.
Sort source data according to the final score Select top-p_src percent source data to construct D'_s.
return Selected source subset D'_s.
```

The domain discrepancies between D_s' and D_t are smaller compared to those between D_s and D_t. Consequently, training the detector on D_s' can improve the performance on the target domain without accessing the target labels. In addition, after training on the target-like source data, the model can extract more accurate target domain features, which is beneficial for the subsequent hybrid active target sampling strategy.

C. Hybrid Active Target Sampling Strategy

Although the performance on the target domain can be boosted using the hybrid active source strategy, the detection accuracy is still largely behind the fully supervised approach. To further adapt the model to the target domain, we propose a hybrid active target sampling strategy that comprehensively considers the rare class problem, instance-level diversity, and scene-level diversity to select a small subset of target data. In the following part, we will detail the hybrid active target sampling strategy.

1) Rare-Class-Aware Sampling Criterion: As shown in Table I, the instance number of some classes (i.e., pedestrian and cyclist) is much smaller than that of other classes, which can be referred to as rare classes. Such a long-tail distribution of categories makes the model easily overfit to the dominant classes, especially in active learning scenarios, where only a small amount of labeled target data can be accessed. Besides, it can be seen in Table II that the results of random sampling even fall behind the source-only results on the rare classes. For example, the source only result on cyclist is 53.98% / 50.11% in $\mathrm{AP_{BEV}}$ / $\mathrm{AP_{3D}}$ and when fine-tuning with randomly sampled 1% frames, the performance decreases to 40.65% / 35.89%. This is because a large proportion of frames do not contain any instances of rare classes, such as highways without sidewalks. According to the statistics, there are 86.15% frames in the KITTI training set that contain no cyclist, and 74.27% frames contain no pedestrian. As a result, it is easy to sample frames without rare classes by simply applying random sampling which severely degrades the performance on these classes.

Based on the observation, it is crucial to pay more attention to the rare classes when sampling target data. To this end, we design a rare-class-aware sampling criterion to select frames with more rare-class instances. In detail, similar to the similarity bank mentioned in Section III-B2. We build a rare-class memory bank B_r which contains the prototypes of rare classes (*e.g.*, pedestrian and cyclist) which is initialized by the rare-class prototypes in B_t as mentioned in Section III-B2 and updated in the training process. The updating process of B_r is similar to that of B_t. Specifically, assuming that k rounds of target domain sampling have been completed, given sampled target data $(x_i^t, y_i^t) \in D'_{t,k}$ where $D'_{t,k}$ is the labeled target set after k rounds of sampling. The RoI features of rare classes $\mathbf{r}_i^t = \{r_{i,j}^t\}_{j=1}^{N_{RoI}}$, their corresponding predicted class $\mathbf{c}_i^t = \{c_{i,j}^t\}_{j=1}^{N_{RoI}}$ and the overlaps with ground-truth bounding boxes $\mathbf{o}_i^t = \{o_{i,j}^t\}_{j=1}^{N_{RoI}}$ can be obtained by the model. To obtain more accurate prototypes, we use the overlap of each RoI to reweight the RoI features as follows:

$$p_{r,i}^c = \mathrm{Softmax}(\mathbf{o}_i^{t,c})(\mathbf{r}_i^{t,c})^{\mathrm{T}}, \tag{7}$$

where $\mathbf{o}_i^{t,c}$ and $\mathbf{r}_i^{t,c}$ denote overlaps and RoI features of the class c and $(\cdot)^{\mathrm{T}}$ represents matrix transpose operation. Then, the rare-class similarity bank B_t can be updated by:

$$p_r^c \leftarrow \beta p_r^c + (1-\beta) p_{r,i}^c, \tag{8}$$

where β is a hyperparameter and p_r^c is the rare-class prototype of class c in B_r.

To sample frames with more abundant rare-class instances, we calculate the rare class similarity between RoIs of unlabeled target data and B_r. In detail, in the k-th sampling round, the RoI features of unlabeled target data $x_i^t \in D_t / D'_{t,k-1}$ are first filtered by Non-Maximum Suppression (NMS). Then the similarity can be obtained by calculating the cosine similarity between the RoI features and prototypes in B_r and the rare-class similarity score can be calculated by summing the cosine similarity. The formula can be written as follows:

$$s_i^{rare} = \sum_{c=1}^{N_{rare}} \sum_{j=1}^{N_{RoI}} \frac{p_r^c \cdot r_{i,j}^{t,c}}{\|p_r^c\| \|r_{i,j}^{t,c}\|}, \tag{9}$$

where N_{rare} and N_{RoI} represent the number of rare classes and the number of RoIs, $p_r^c \in B_r$ and $r_{i,j}^{t,c}$ denotes the j-th RoI feature of class c in i-th frame. The s_i^{rare} is then served as a metric to select target domain data.

2) Intra-Frame Diversity Criterion: As shown in Table III and Table IV, the existing 2D ADA methods fail to achieve satisfactory performance. One key reason is that they tend to sample similar frames, since adjacent frames in the road scenes often yield similar scores under active learning criteria. This results in redundant annotations and increases the risk of the model overfitting to a limited number of instances or frames. Consequently, ensuring diversity in both sampled instances and frames is crucial for ADA in 3D object detection tasks.

As shown in Fig. 5, the point cloud shapes of objects within the same category exhibit significant variations due to factors such as distance from the LiDAR sensor, occlusion by other objects, and differences in object shapes. The network will overfit if sampled frames contain highly similar instances. For example, the model will overfit to the instance with a more dense point cloud if most instances in sampled frames are close to the ego car and consequently struggle to detect objects at a distance. Based on the observation, we propose an intra-frame diversity criterion that evaluates the diversity of RoI features within each class to sample instances with diversity which will lead to better model performance in the target domain.

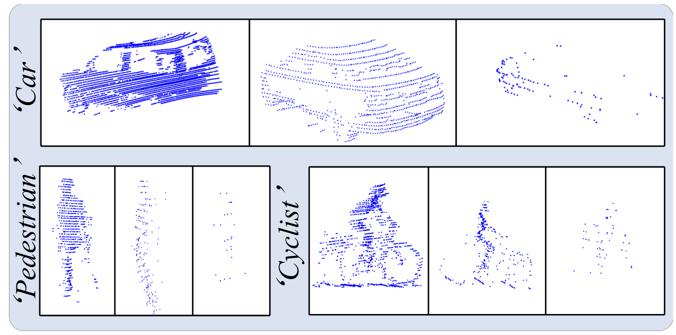


Fig. 5. The Diversity of instances in the KITTI dataset which is caused by the different distances from the LiDAR sensor and the occlusion of other objects.

In detail, given the RoI features $\mathbf{r}_i^t = \{r_{i,j}^t\}_{j=1}^{N_{RoI}}$, their corresponding classes $\mathbf{c}_i^t = \{c_{i,j}^t\}_{j=1}^{N_{RoI}}$ and confidence score $\mathbf{a}_i^t = \{a_{i,j}^t\}_{j=1}^{N_{RoI}}$ preprocessed by NMS. The RoI features are then filtered by a threshold of the confidence score τ^c, which is simply set to 0.7 for cars and 0.5 for pedestrians and cyclists. To measure the diversity of RoI features, we calculate the cosine similarity between the RoI features of class c and the prototype of c. Specifically, the prototype of each class can be calculated by weighting RoI features by confidence scores as follows:

$$p_{d,i}^c = \mathrm{Softmax}(\mathbf{a}_i^{t,c})(\mathbf{r}_i^{t,c})^{\mathrm{T}}, \tag{10}$$

where $p_{d,i}^c$ denotes the prototype of class c in i-th frame. After obtaining the prototypes, we then calculate the similarity between $p_{d,i}^c$ and $\{r_{i,j}^{t,c}\}_{j=1}^{N_{RoI}^c}$. A high similarity means the diversity is small in this frame. The final intra-frame diversity score can be calculated by summing the diversity of each class, which can be formulated as the following equation:

$$s_i^{intra} = \sum_{c=1}^{N_c} \left(1 - \frac{1}{N_{RoI}^c} \sum_{j=1}^{N_{RoI}^c} \frac{p_d^c \cdot r_{i,j}^{t,c}}{\|p_d^c\| \|r_{i,j}^{t,c}\|}\right), \tag{11}$$

where s_i^{intra} denotes the intra-frame diversity score of the i-th frame.

3) Inter-Frame Diversity-Based Hybrid Active Target Sampling Criterion: As mentioned above, the diversity between frames is also important when designing the active sampling strategy to avoid overfitting to similar frames (*e.g.*, a sequence in an autonomous driving scene). To address this, we propose an inter-frame diversity-based hybrid active target sampling strategy to sample frames based on the inter-frame diversity and meanwhile considering the rare-class-aware similarity and intra-frame diversity as proposed in Section III-C1 and

TABLE III
RESULTS ON DIFFERENT ADAPTATION SCENARIOS UNDER 1% ANNOTATION BUDGET USING SECOND-IOU AS OUR BASELINE DETECTOR. WE REPORT THE MEAN AP OF ALL CLASSES IN THE LAST ROW AND THE BEST ADAPTATION RESULTS ARE MARKED IN **BOLD**.

Task	Method	Setting	SECOND AP_{BEV} / AP_{3D}			
			Car	Pedestrian	Cyclist	Mean
Waymo→KITTI	Source Only	-	51.48 / 19.78	40.80 / 31.26	47.63 / 35.45	46.64 / 28.83
	SN [53]	-	76.61 / 54.14	52.48 / 48.20	34.56 / 32.74	54.55 / 45.03
	ST3D [7]	UDA	77.62 / 49.24	44.45 / 42.04	47.74 / 42.95	56.60 / 44.70
	ST3D++ [23]	UDA	77.68 / 50.03	49.09 / 46.19	51.50 / 47.70	59.42 / 47.97
	REDB [10]	UDA	80.37 / 54.12	51.01 / 48.20	52.05 / 47.97	61.14 / 50.10
	SSDA3D [24]	SSDA	76.05 / 60.44	50.45 / 45.96	51.74 / 47.57	59.41 / 51.32
	TQS [26]	ADA	80.45 / 64.75	50.98 / 47.97	51.43 / 47.05	60.95 / 53.25
	CLUE [27]	ADA	82.23 / 65.85	51.46 / 47.56	**53.71** / 50.57	62.46 / 54.66
	Bi3D [30]	ADA	83.56 / 69.89	52.31 / 48.57	52.66 / 49.21	62.84 / 55.89
	Bi3D++	ADA	**85.62** / **71.07**	**54.98** / **50.41**	53.68 / **51.47**	**64.75** / **57.65**
	Oracle	-	83.29 / 73.45	46.64 / 41.33	62.92 / 60.32	64.28 / 58.37
Waymo→nuScenes	Source Only	-	30.64 / 17.18	1.81 / 0.58	0.97 / 0.88	11.14 / 6.22
	SN [53]	-	29.56 / 17.78	1.33 / 1.07	2.92 / 2.61	11.27 / 7.15
	ST3D [7]	UDA	28.42 / 17.83	1.64 / 1.39	4.01 / 3.54	11.36 / 7.59
	ST3D++ [23]	UDA	28.87 / 19.15	1.82 / 1.58	4.09 / 3.74	11.60 / 8.16
	REDB [10]	UDA	30.12 / 18.56	2.47 / 2.14	6.56 / 5.19	13.05 / 8.63
	SSDA3D [24]	SSDA	37.21 / 23.02	13.62 / 9.67	7.70 / 6.43	19.51 / 13.04
	TQS [26]	ADA	33.31 / 20.07	10.26 / 7.17	5.54 / 4.22	16.37 / 10.48
	CLUE [27]	ADA	32.58 / 19.79	10.34 / 7.45	6.87 / 5.65	16.59 / 10.96
	Bi3D [30]	ADA	36.43 / 23.96	13.77 / 10.20	6.23 / 5.46	18.81 / 13.20
	Bi3D++	ADA	**37.51** / **24.52**	**14.24** / **10.67**	**8.62** / **7.93**	**20.12** / **14.37**
	Oracle	-	51.88 / 34.87	25.24 / 18.92	15.06 / 11.73	30.73 / 21.84
nuScenes→KITTI	Source Only	-	39.15 / 7.65	21.54 / 16.87	6.31 / 2.44	22.33 / 8.98
	SN [53]	-	56.08 / 28.67	23.05 / 16.84	5.11 / 2.32	28.08 / 15.94
	ST3D [7]	UDA	71.50 / 48.09	22.64 / 17.61	7.86 / 5.20	34.00 / 23.64
	ST3D++ [23]	UDA	69.90 / 44.62	24.11 / 18.20	10.14 / 6.39	33.75 / 23.07
	REDB [10]	UDA	74.23 / 51.31	25.95 / 18.38	13.82 / 8.64	38.00 / 26.11
	SSDA3D [24]	SSDA	75.63 / 52.36	24.87 / 18.21	15.63 / 10.47	38.71 / 27.01
	TQS [26]	ADA	79.57 / 59.68	26.18 / 21.94	14.33 / 9.97	40.02 / 30.53
	CLUE [27]	ADA	79.55 / 58.65	30.65 / 24.84	16.63 / 13.86	42.27 / 32.45
	Bi3D [30]	ADA	76.66 / 57.90	30.17 / 24.75	20.49 / 16.84	42.44 / 33.16
	Bi3D++	ADA	**80.10** / **63.03**	**33.18** / **27.22**	**22.83** / **18.77**	**45.37** / **36.34**
	Oracle	-	83.29 / 73.45	46.64 / 41.33	62.92 / 60.32	64.28 / 58.37

Section III-C2. More specifically, we use the distance between scene-level representation to measure the diversity of frames and cluster the unlabeled frames into b_k groups where b_k is the budget of the k-th sampling round. Then, by comparing the rare-class similarity and intra-frame diversity in each group, the final $\Delta D'_{t,k}$ can be obtained and further annotated where $\Delta D'_{t,k}$ is the set of selected target data in the k-th sampling round.

In detail, given $x_i^t \in D_t \setminus D'_{t,k-1}$, the scene-level representation f_i^{sc} can be obtained by applying the global average pooling on the output of the final convolution layer of the domain discriminator. Then we cluster all frames in $D_t \setminus D'_{t,k-1}$ into b_k groups according to the f_i^{sc} by calculating the distance between f_i^{sc} and each group prototype $\mathbf{p}^{sc} = \{p_g^{sc}\}_{g=1}^{b_k}$. The prototype of each group is initialized as empty and updated by the following formula:

$$\hat{p}_{m,n}^{sc} = \frac{\text{num}(G_m) \cdot p_m^{sc} + \text{num}(G_n) \cdot p_n^{sc}}{\text{num}(G_m) + \text{num}(G_n)}, \tag{12}$$

where G_m and G_n represent the m-th and n-th group respectively, $\hat{p}_{m,n}^{sc}$ is the updated prototype of combining G_m and G_n. The detailed operation can be found in Algorithm 2. After obtaining the selected unlabeled data $\Delta D'_{t,k}$, the annotation of $\Delta D'_{t,k}$ can be obtained by the oracle (usually the human annotator) and $D'_{t,k}$ can be acquired by merging $D'_{t,k-1}$ and $\Delta D'_{t,k}$.

D. Cross-Domain GT Sampling and Class-Balance Re-Sampling

To further alleviate the rare-class issue, inspired by [59], [60], we use cross-domain GT sampling and class-balance re-sampling to further improve the model's performance on rare classes.

Cross-domain GT Sampling: As mentioned before, the categories in road scenes present a long-tail distribution, as a result, the sampled frames contain a small number of rare-class instances. Fortunately, there exists a relatively large number of

TABLE IV
RESULTS ON DIFFERENT ADAPTATION SCENARIOS UNDER 1% ANNOTATION BUDGET USING PV-RCNN AS OUR BASELINE DETECTOR. WE REPORT THE MEAN AP OF ALL CLASSES IN THE LAST ROW AND THE BEST ADAPTATION RESULTS ARE MARKED IN **BOLD**.

Task	Method	Setting	PV-RCNN AP_{BEV} / AP_{3D}			
			Car	Pedestrian	Cyclist	Mean
Waymo→KITTI	Source Only	-	58.25 / 10.48	51.58 / 49.06	51.97 / 49.10	53.93 / 36.21
	SN [53]	-	74.09 / 59.61	53.86 / 50.30	48.91 / 46.57	58.95 / 52.16
	ST3D [7]	UDA	83.93 / 63.25	47.72 / 44.20	52.40 / 48.68	61.35 / 52.04
	ST3D++ [23]	UDA	84.02 / 64.74	52.25 / 49.97	56.54 / 52.45	64.27 / 55.72
	ReDB [10]	UDA	85.25 / 68.37	54.46 / 52.88	57.65 / 53.26	65.79 / 58.17
	SSDA3D [24]	SSDA	82.65 / 71.45	52.70 / 49.50	53.88 / 48.76	63.07 / 56.57
	TQS [26]	ADA	84.04 / 71.77	58.24 / 56.27	58.15 / 56.15	66.81 / 61.40
	CLUE [27]	ADA	85.38 / 72.79	61.24 / 58.04	60.79 / 58.05	69.14 / 62.96
	Bi3D [30]	ADA	86.36 / 71.84	61.64 / 59.51	61.91 / 60.01	69.97 / 63.79
	Bi3D++	ADA	**86.57** / **77.55**	**64.23** / **62.18**	**62.34** / **60.14**	**71.04** / **66.62**
	Oracle	-	88.98 / 82.50	54.13 / 49.96	73.65 / 70.69	72.25 / 67.71
Waymo→nuScenes	Source Only	-	28.08 / 16.74	9.05 / 6.86	0.70 / 0.57	12.61 / 8.06
	SN [53]	-	28.80 / 17.20	10.72 / 7.85	0.70 / 0.64	13.41 / 8.56
	ST3D [7]	UDA	30.21 / 18.62	10.15 / 7.26	1.33 / 1.17	13.90 / 9.02
	ST3D++ [23]	UDA	31.58 / 19.86	10.64 / 7.53	1.97 / 1.72	14.73 / 9.70
	ReDB [10]	UDA	31.67 / 20.55	13.39 / 10.27	3.52 / 2.54	16.19 / 11.12
	SSDA3D [24]	SSDA	42.30 / 27.82	26.22 / 19.32	3.32 / 2.90	23.95 / 16.68
	TQS [26]	ADA	37.67 / 24.19	22.37 / 15.84	2.28 / 1.96	20.77 / 14.00
	CLUE [27]	ADA	38.14 / 24.86	23.66 / 16.75	3.38 / 2.59	21.73 / 14.73
	Bi3D [30]	ADA	41.69 / 27.54	26.47 / 19.15	5.59 / 4.38	24.58 / 17.02
	Bi3D++	ADA	**44.57** / **29.47**	**28.73** / **22.07**	**7.38** / **6.04**	**26.89** / **19.19**
	Oracle	-	51.88 / 34.87	25.24 / 18.92	15.06 / 11.73	30.73 / 21.84
nuScenes→KITTI	Source Only	-	59.49 / 25.67	17.77 / 14.01	11.55 / 9.61	29.60 / 16.43
	SN [53]	-	70.65 / 42.84	19.56 / 16.25	11.63 / 9.84	33.95 / 22.98
	ST3D [7]	UDA	72.10 / 53.03	23.68 / 21.04	12.88 / 10.06	36.22 / 28.04
	ST3D++ [23]	UDA	73.39 / 55.97	23.16 / 20.31	12.61 / 9.86	36.39 / 28.71
	ReDB [10]	UDA	76.26 / 58.49	25.28 / 22.77	13.54 / 10.29	38.36 / 30.52
	SSDA3D [24]	SSDA	79.55 / 65.85	33.68 / 26.42	15.46 / 13.08	42.90 / 35.12
	TQS [26]	ADA	82.25 / 71.63	35.37 / 27.29	13.05 / 11.33	43.56 / 36.75
	CLUE [27]	ADA	82.73 / 69.33	38.69 / 31.47	16.03 / 13.57	45.82 / 38.12
	Bi3D [30]	ADA	82.44 / 71.95	38.78 / 31.98	20.66 / 17.47	47.29 / 40.47
	Bi3D++	ADA	**84.76** / **72.92**	**40.20** / **33.43**	**26.96** / **24.55**	**50.64** / **43.63**
	Oracle	-	88.98 / 82.50	54.13 / 49.96	73.65 / 70.69	72.25 / 67.61

such rare-class instances compared to sampled frames. To this end, we design cross-domain GT sampling to further improve the detection accuracy on rare classes. In detail, given the GT labels and their corresponding point cloud in the source domain, we can randomly sample the instances of the rare classes and paste the point clouds onto the sampled target domain frames. Note that if there is an overlap between the sampled instance and instances in the current frame, the sampled instance will not be pasted onto the current frame. However, there still exist domain discrepancies between instances from the source and target domains. To mitigate this, we simply turn off the cross-domain GT sampling in the last 5 epochs.

Class-balance Re-sampling: Inspired by widely used class-balance sampling in the nuScenes dataset, we use the same strategy to further enhance the model's ability to detect rare categories. Specifically, given the instance number of each class in sampled frames, the proportion of each class can be calculated and then we oversample the rare classes and undersample the dominant classes according to these proportions.

E. Overall Objectives and Bi-Domain Sampling Strategies

To adapt the model to the target domain, our proposed Bi3D++ can be divided into the following four steps.

1) Pre-training on the source domain: The detector is first pre-trained on the labeled source domain to learn knowledge of classification and localization which is supervised by the following equation:

$$\mathcal{L}_{detector} = \mathcal{L}_{rpn} + \mathcal{L}_{rcnn}, \tag{13}$$

where $\mathcal{L}_{rpn}$ and $\mathcal{L}_{rcnn}$ represent loss of RPN and IoU head respectively.

Algorithm 2: Diversity-based Hybrid Active Target Strategy.

Input: Unlabeled target dataset $D_t/D'_{t,k-1}$ and the annotation budget of the k-th sampling round.
Output: The selected target set $\Delta D'_{t,k}$.
Initialize each group $\mathbf{G} = \emptyset$ and prototypes of each group $\{p_g^{sc}\}_{g=1}^{b_k} = \emptyset$.
for x_i^t *in* $D_t/D'_{t,k-1}$ **do**
  Calculate the scene-level representation f_i^{sc}.
  Calculate the rare-class similarity score s_i^{rare} and intra-frame diversity score s_i^{intra} using Eq. (9) and Eq. (11).
  if $i < b_k$ **then**
    Update $G_i = \{x_i^t\}$, $p_i^{sc} = f_i^{sc}$.
  else
    Calculate the cosine similarity $\alpha_{f_i^{sc},\mathbf{p}^{sc}}$ between f_i^{sc} and $\mathbf{p}$.
    Calculate the cosine similarity $\alpha_{\mathbf{p}^{sc}}$ of prototypes in $\mathbf{p}^{sc}$.
    if $\max(\alpha_{f_i^{sc},\mathbf{p}^{sc}}) < \min(\alpha_{\mathbf{p}^{sc}})$ **then**
      Merge the most similar banks G_m and G_n and the corresponding prototypes p_m^{sc} and p_n^{sc} using Eq. (12).
      Update $\mathbf{G} = \mathbf{G} \cup x_i^t$, $\mathbf{p^s c} = \mathbf{p^{sc}} \cup f_i^{sc}$.
    else
      Merge f_i^{sc} into G_m, where the corresponding prototype p_m^{sc} is most similar to f_i^{sc}.
for j *in* $[1, b_k]$ **do**
  Select data with max score $s_i^{rare} + s_i^{intra}$ to construct $\Delta D'_{t,k}$.
return Selected target subset $\Delta D'_{t,k}$.

2) Training the domain discriminator: In this step, the domain discriminator is trained only using (3). Note that the parameters of the detector is frozen in this step and only the parameter of the domain discriminator is optimized.

3) Active sampling source domain: In this step, we select target-like source data D'_s by the proposed hybrid active sampling strategy as shown in Algorithm 1, and then the detector is fine-tuned on D'_s using (13) to improve the performance on the target domain.

(4) Active sampling target domain: Finally, utilizing the proposed hybrid target sampling strategy Algorithm 2, we can sample D'_t, and then the model is trained on both D'_s and D'_t using (13).

IV. Experiments and Results

A. Experimental Setup

1) Datasets Description and Cross-Domain Settings: We conduct experiments on three widely-used 3D object detection datasets including KITTI [6], Waymo [5] and nuScenes [58].

KITTI: KITTI dataset is one of the most popular datasets for autonomous driving which is collected by a 64-beam LiDAR in Germany. Since only annotations of the field of view of the front (FOV) camera are provided, following [7], we only consider the points inside the range in the test phase.

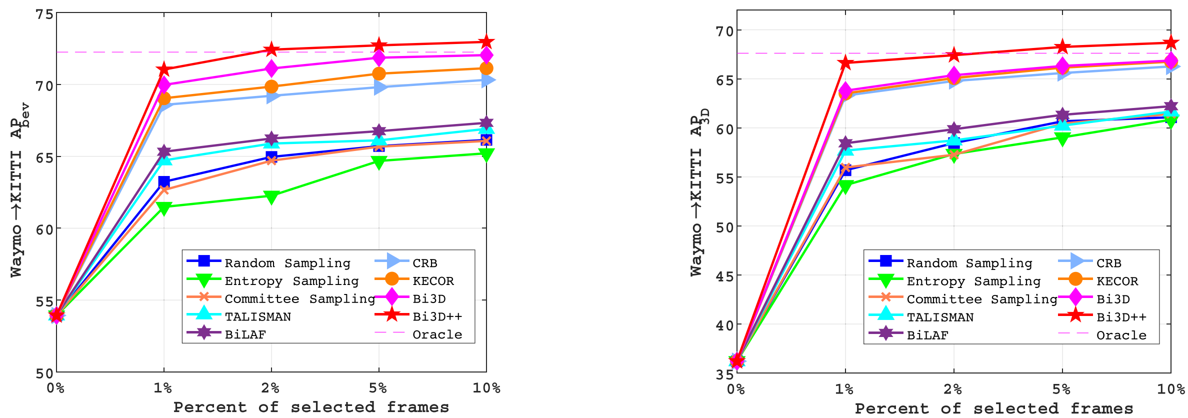
Waymo: Waymo Open Dataset provides point cloud data from a 64-beam LiDAR which is composed of 1,000 sequences. We follow [23] and uniformly sample one-fifth data from its training set to train the source model when Waymo is regarded as the source domain.

nuScenes: The nuScenes dataset is gathered from a 32-beam LiDAR in Singapore and Boston, USA. It consists of 28,130 training samples and 6,019 validation samples. It contains data at various times and weather conditions.

Cross-domain Settings: Following [10], we conduct experiments on three cross-domain settings including Waymo → KITTI, Waymo → nuScenes, nuScenes → KITTI. These three cross-domain settings include cross-location (*e.g.*, Waymo → KITTI) and cross-sensor (*e.g.*, nuScenes → KITTI).

2) Comparison Methods and Evaluation Metrics: Comparison Methods: To verify the effectiveness of Bi3D++, we compare the performance on multiple cross-domain settings with 1) *State-of-the-art UDA works* including ST3D [7], ST3D++ [23], ReDB [10]. ST3D is a self-training-based method that continuously updates the pseudo-labels by designing a quality-aware triplet memory bank. ST3D++ extends ST3D by analyzing the impact of noisy pseudo labels and introducing a more accurate pseudo-label generation method. ReDB first explores UDA for 3D object detection on multiple categories. 2) *Reproduced commonly-used 2D ADA works* including TQS [26] and CLUE [27]. TQS designs a sampling strategy by comprehensively considering transferable committee, transferable uncertainty, and transferable domainness. CLUE samples target domain data by introducing a weighted-cluster method. 3) *Existing 2D and 3D active learning methods* including query-by-committee [39], query-by-uncertainty [40], TALISMAN [45], BiLAF [44], CRB [46], KECOR [61]. Query-by-committee and query-by-uncertainty are widely used active learning methods and lots of following active learning works are inspired by them. CRB is a pioneer in exploring active learning in 3D object detection. KECOR uses an information-theoretic-based criterion to effectively select data. (4) *SSDA methods* (i.e., SSDA3D [24]). SSDA3D introduces a novel SSDA framework that is composed of inter-domain adaptation and intra-domain generalization. Further, we report the source only, SN [53], and the oracle results. Source only means directly evaluating the target domain performance on the model which is pre-trained on the source domain. SN first analyzes the domain discrepancies in 3D object detection and introduces statistical normalization to handle the different distribution of box size between the source and target domain which needs the statistics of the target domain. Oracle denotes the model's performance trained on a fully annotated target domain.

Evaluation Metrics: For a fair comparison with previous works [10], [30], we use the KITTI evaluation metric to evaluate performance on three common classes in autonomous driving scenarios, including car (vehicle in Waymo dataset), pedestrian,



(a) AP_{BEV} results. (b) AP_{3D} results.

Fig. 6. Results under different annotation budgets. We report the results on the Waymo $\rightarrow$ KITTI scenario using PV-RCNN as our baseline detector.

and cyclist (in nuScenes dataset, cyclist is obtained by merging bicycle and motorcycle). We report the average precision in both BEV and 3D (i.e., AP_{BEV} / AP_{3D}) over 40 recall positions, with the IoU threshold 0.7 for car and 0.5 for pedestrian and cyclist.

3) Implementation Details: Following [10], we verify the effectiveness of Bi3D++ using SECOND-IoU as our baseline detector. To demonstrate our proposed method can be combined with different detectors, following [7], we further conduct experiments on PV-RCNN [28]. The source model is pre-trained on the source domain for three classes using the random object scaling (ROS) augmentation method. For a fair comparison, we only use the coordinate encoding (x, y, z) of the raw point clouds as the detector input and set the voxel size to (0.1 m, 0.1 m, 0.15 m) on all datasets. The hyperparameters α and β are set to 0.99. We use Adam optimizer with one-cycle scheduler and set the learning rate to 1×10^{-3}. The commonly-used data augmentation methods (*e.g.*, random world flip, random world rotation, random world scaling) are utilized in our experiments. In the stage of active sampling source domain, we first select the target-like data from the source domain in the initial training epoch and fine-tune the detector for the following 15 epochs. In the stage of active sampling target domain, we mainly consider the situation that the annotation budget B is equal to 1% in our main results. Our code is built on 3DTrans [62]. Note that we use techniques discussed in Section III-D by default unless otherwise specified.

B. Main Results

As shown in Table III and Table IV, our proposed method can be combined with different detectors (*e.g.*, SECOND-IoU and PV-RCNN) and can deal with multiple domain discrepancies such as location and sensor. In the following part, we show the effectiveness of our proposed method by comparing it with ADA, SSDA, and UDA methods, respectively.

Comparison with ADA works: Since ADA for 3D detection is still under-explored, for a fair comparison, we first compare our proposed Bi3D++ with previous 2D ADA works under the same cross-domain settings (i.e., sampling 1% target data to annotate). As shown in Table III and Table IV, we can observe that existing 2D ADA works cannot achieve satisfactory results and even fall behind some UDA works, which is because these methods often sample similar frames and ignore rare classes while sampling, resulting in overfitting to similar frames and a single category. Besides, they have not considered the long-tail distribution in the target domain. For example, TQS [26] only achieves 60.95% / 53.59% in mean AP_{BEV} / AP_{3D} on Waymo $\rightarrow$ KITTI scenario using SECOND as baseline detector and even falls behind the UDA method on some classes (i.e., pedestrian and cyclist compared to ReDB [10]). In contrast, our Bi3D++ fully considers the instance-level and scene-level diversity as well as the rare-class problem, thus selecting more valuable target data and largely outperforms 2D ADA methods (*e.g.*, +3.80% / +4.06% AP_{BEV} / AP_{3D} on Waymo $\rightarrow$ KITTI scenario compared to TQS under the same setting), further showing that our Bi3D++ is more suitable for 3D detection tasks and for multiple classes domain adaptation.

Comparison with SSDA works: Besides, we compare our proposed method with SSDA3D [24] where we randomly sample the same number of target data (i.e., 1%) to perform the semi-supervised manner. Note that the SSDA and ADA are orthogonal which will be further analyzed in Section IV-C2. It can be seen in Table III and Table IV that SSDA3D cannot achieve satisfactory results, especially in rare classes due to only a few rare-class instances exist in randomly sampled frames as mentioned in Section III-C1. For example, SSDA3D only achieves 50.45% / 45.96% and 51.74% / 47.57% in pedestrian and cyclist on Waymo $\rightarrow$ KITTI scenario, respectively when using SECOND as the baseline detector, while our Bi3D++ can achieve 54.98% / 50.41% and 53.68% / 51.47% using the same number of labeled target data. The results further demonstrate that our proposed methods can sample more rare-class instances and avoid overfitting to the dominant class.

Comparison with UDA works: Most existing works tackle cross-domain 3D object detection tasks in the UDA manner. To demonstrate the advantage of active learning, we further compare it to state-of-the-art UDA works. It can be seen from

TABLE V
THE RESULT OF COMBINING WITH DIFFERENT DA METHODS (I.E., WDA, UDA, SSDA). WE CONDUCT THE EXPERIMENTS ON WAYMO → KITTI SCENARIOS USING PV-RCNN AS OUR BASELINE DETECTOR.

Method	Setting	Car	Pedestrian	Cyclist	Mean
Bi3D++	ADA	86.57 / 77.55	64.23 / 62.18	62.34 / 60.14	71.04 / 66.62
SN [53]	WDA	74.09 / 59.61	53.86 / 50.30	48.91 / 46.57	58.95 / 52.16
SN [53] + Bi3D++	WDA + ADA	86.92 / 77.95	64.54 / 62.45	62.92 / 60.88	71.46 / 67.09
ReDB [10]	UDA	85.25 / 68.37	55.46 / 52.88	57.65 / 53.26	66.11 / 64.45
ReDB [10] + Bi3D++	UDA + ADA	87.23 / 78.12	65.65 / 63.04	63.75 / 61.98	72.21 / 67.71
SSDA3D [24]	SSDA	82.65 / 71.45	52.70 / 49.50	53.88 / 48.76	63.07 / 56.57
SSDA3D [24] + Bi3D++	SSDA + ADA	87.45 / 78.23	65.41 / 62.82	63.48 / 61.61	72.11 / 67.55

TABLE VI
ABLATION STUDIES OF EACH COMPONENT. INS., DOM. REPRESENT INSTANCE-LEVEL SIMILARITY AND SCENE-LEVEL DOMAINNESS IN THE ACTIVE SOURCE SAMPLING STRATEGY, RESPECTIVELY. RARE., INTRA. AND INTER. REPRESENT THE RARE-CLASS SIMILARITY CRITERION, THE INTRA-FRAME DIVERSITY CRITERION, AND THE INTER-FRAME DIVERSITY CRITERION IN THE ACTIVE SAMPLING TARGET DOMAIN STRATEGY. THE EXPERIMENTS ARE CONDUCTED ON WAYMO → KITTI SCENARIO USING PV-RCNN AS THE BASELINE DETECTOR.

Source		Target			AP_{BEV} / AP_{3D}			
Dom.	Ins.	Rare.	Intra.	Inter.	Car	Pedestrian	Cyclist	Mean
-	-	-	-	-	76.34 / 45.76	56.79 / 54.28	51.11 / 45.46	61.41 / 48.50
✓	-	-	-	-	83.37 / 62.49	59.90 / 58.06	60.39 / 58.35	67.56 / 60.24
-	✓	-	-	-	79.62 / 48.03	58.22 / 55.65	54.06 / 49.86	63.96 / 51.18
✓	✓	-	-	-	83.72 / 68.03	60.58 / 58.61	60.48 / 58.61	68.26 / 61.75
✓	✓	✓	-	-	85.87 / 74.02	63.90 / 61.48	60.71 / 58.72	70.16 / 64.74
✓	✓	-	✓	-	82.96 / 71.43	63.41 / 60.61	61.20 / 59.98	69.19 / 64.00
✓	✓	✓	✓	-	85.96 / 74.21	63.93 / 61.82	**62.89** / **60.55**	70.92 / 65.52
✓	✓	✓	-	✓	86.22 / 75.24	63.68 / 61.23	61.34 / 59.45	70.41 / 65.30
✓	✓	-	✓	✓	86.12 / 73.58	63.60 / 61.13	61.78 / 60.10	70.05 / 64.93
✓	✓	✓	✓	✓	**86.57** / **77.55**	**64.23** / **62.18**	62.34 / 60.14	**71.04** / **66.62**

Table III and Table IV that Bi3D++ can surpass all UDA methods with a large margin by only using 1% labeled target data. For example, compared to the SOTA UDA method (i.e., ReDB [10]) which already considers multi-class problems, Bi3D++ can also achieve 64.75% / 57.65% on Waymo → KITTI scenario which surpasses ReDB using only 1% labeled target data. When compared to works designed for a single class (*e.g.*, ST3D), the improvements on rare classes are +8.37% and +8.52% AP_{3D} which further shows that the rare-class problem is essential in such a task. Furthermore, on a more difficult cross-domain setting (i.e., Waymo → nuScenes), Bi3D++ can achieve 20.12% / 14.08%, with +7.07% / +5.45% improvement, which largely reduces the performance gap between source only and oracle.

C. Insight Analysis

1) Results of Changing Annotation Budget: In this section, we compare our proposed method with random sampling and commonly used active learning methods, including both 2D (i.e., query-by-committee, query-by-uncertainty, TALISMAN [45], and BiLAF [44]) and 3D (i.e., CRB [46], KECOR [61]) works. We set the annotation budget to 1%, 2%, 5%, and 10% and conduct the experiments on Waymo → KITTI scenarios. Note that for a fair comparison with CRB and KECOR which are instance-level sampling methods, we first calculate the sampling instance number of Bi3D++ and then sample the same number of objects using CRB and KECOR. The trends of AP_{BEV} and AP_{3D} are shown in Fig. 6. Bi3D++ can consistently outperform random sampling and all compared active learning methods under all annotation budgets, which further shows the effectiveness of our proposed methods. Besides, as the annotation budget increases, the performance of Bi3D++ can be continuously improved, and when the annotation budget reaches 5%, the performance on the target domain even surpasses the oracle, which is trained on 100% of annotated target data on Waymo → KITTI scenario.

Although the above-mentioned active learning methods achieve success in the active learning paradigm, they cannot get satisfactory results when domain discrepancies exist. The main reasons can be summarized as follows: (1) Bi3D++ fully considers domain discrepancies in cross-domain settings, for example, the hybrid active source sampling strategy can narrow the distribution gap between the source and target domain and thus reduce the domain discrepancies. However, existing active learning methods do not consider domain discrepancies, for example, in CRB [46], the need to measure the prototype between the labeled set and the unlabeled set will lead to inaccurate sampling in the first sampling round since the labeled data are from the source domain. (2) Bi3D++ fully considers the problems that exist in autonomous driving scenarios (i.e., the long-tail

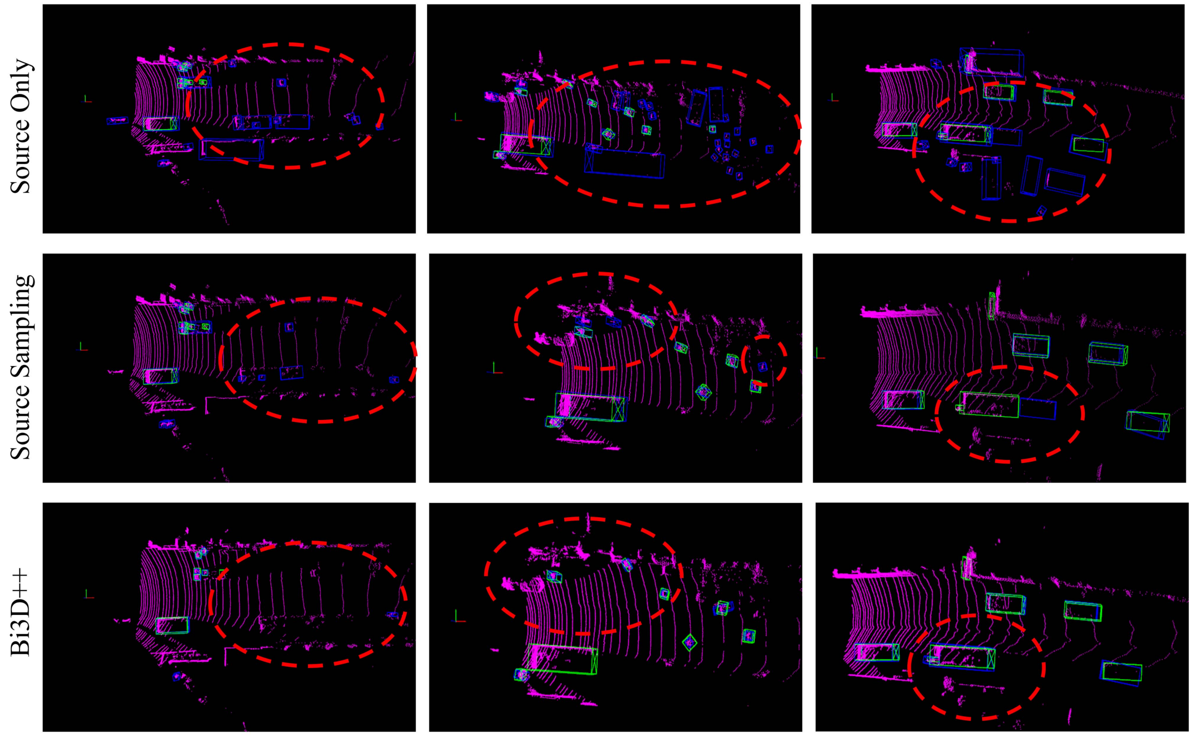


Fig. 7. Qualitative results of each stage. Bi3D++ denotes using both source and target sampling strategies. The blue boxes represent the prediction results of the corresponding model, while the green boxes denote the ground truth. The red circles highlight the specific instances where the "source-only" model made an incorrect prediction, which was then successfully corrected by our proposed Bi3D++ model.

distribution of different categories, the existence of similar frames and instances in AD data. The hybrid active target sampling strategy comprehensively tackles the problem of long-tail distribution of classes and redundant sampling caused by similar frames and instances. In contrast, existing active learning methods (*e.g.*, query-by-committee, query-by-uncertainty) cannot handle such problems, resulting in relatively low performance.

2) Results of Combining With Other Domain Adaptation Methods: As mentioned before, ADA is orthogonal to SSDA methods. The SSDA methods focus on designing algorithms to make better use of labeled target data, which is mainly from the perspective of training algorithms while ADA works concentrate on selecting the most valuable target domain dat,a which is from the perspective of data. To verify that our proposed method can be combined with the SSDA method and further improve the performance, we conduct the experiments of combining Bi3D++ and SSDA3D as shown in Table V, the performance can be further improved and surpass both Bi3D++ and SSDA3D which is because Bi3D++ can sample diverse target data which tackles the problems caused by random sampling and SSDA3D makes use of the labeled data and the rest of unlabeled target data which further boost the performance.

Bi3D++ is also orthogonal to self-training-based UDA works. In detail, we utilize our proposed method to select 1% target data to obtain annotation and use UDA (i.e., ReDB [10]) methods to generate pseudo labels of the remaining 99% unlabeled target data. It can be observed in Table V that when using ReDB to generate pseudo-labels for the 99% unlabeled target data, the performance can be further improved and reach 72.21% / 67.71%, which largely outperforms ReDB. Besides, Bi3D++ can be combined with the weakly supervised DA method (i.e., SN [53]), which needs the statistical information of the target domain. In detail, we pre-train the source-only model using SN and then use the designed active learning criteria to sample data and train the model on sampled data. As shown in Table V, combining with SN, the performance of Bi3D++ can reach 71.46% / 67.09%, which outperforms Bi3D++. The results demonstrate that Bi3D++ can be flexibly combined with different types of DA methods and further improve the performance, highlighting the flexibility of our proposed method.

TABLE VII
ABLATION STUDIES ON CROSS-DOMAIN GT SAMPLING AND CLASS BALANCE RE-SAMPLING. THE EXPERIMENTS ARE CONDUCTED ON THE WAYMO → KITTI SCENARIO USING PV-RCNN AS THE BASELINE DETECTOR. OURS REPRESENTS USING THE HYBRID BI-DOMAIN ACTIVE SAMPLING STRATEGY.

Method	AP_{BEV} / AP_{3D}
Random	56.35 / 48.28
Random + Cross GT	61.00 / 54.45
Random + Class Balance	58.77 / 50.12
Random + Cross GT + Class Balance	63.23 / 55.68
Ours	69.01 / 64.87
Ours + Cross GT	69.95 / 65.44
Ours + Class Balance	70.76 / 66.51
Ours + Cross GT + Class Balance	**71.04** / **66.62**

3) Ablation Studies: To analyze the contribution of each designed module, we conduct extensive ablation studies on the

Waymo $\rightarrow$ KITTI scenario using PV-RCNN as our baseline detector and we set the annotation budget to 1%. As shown in the first three rows of Table VI, the scene-level domainness can largely improve the performance on the target domain (i.e., +6.15% / +11.74%). This is because we sample target-like source data which have a more similar distribution to the target domain while avoiding interference with the widespread source domain distribution. Further, by utilizing the instance-level similarity criterion, better results (i.e., 68.26% / 61.75%) can be achieved without access to target domain labels since instances in sampled source data can have a similar representation to the target domain instances.

Besides, it can be seen in the lower part of Table VI that we ablate each component in the active target sampling strategy. Firstly, the rare-class similarity criterion can largely improve the performance on rare classes (*e.g.*, +3.32% / +2.87% on pedestrian) because we can acquire frames with more rare-class instances by using such a criterion. The intra-frame diversity can improve the performance of all categories since it can sample frames with more diverse instances (i.e., +0.93% / +2.25% on mean AP_{BEV}/AP_{3D} compared to model obtained after the active source sampling stage) and thus avoiding overfitting to the dominant category. In addition, by using inter-frame diversity to sample frames with scene-level diversity, better results can be obtained, as shown in the last three lines in Table VI and the best results with the improvement of 10.70% / 19.05% compared to the baseline can be obtained when using all three criteria at the same time.

As shown in Table VII, we also conduct ablation studies on cross-domain gt sampling and class-balance re-sampling. It can be observed that both cross-domain gt sampling and class-balance re-sampling can further boost the performance on the target domain since they increase the number of rare-class instances. This is due to 1) the cross-domain GT sampling technique further increases the diversity of the instances and consequently improves the model's generalization ability, and 2) the class-balance re-sampling technique further balances the number of instances of each category (e.g., data with more rare-class instances may be sampled more than once in one epoch) and will improve the performance, especially on rare classes. By comparing random sampling and Bi3D++, we find that more improvement can be obtained by combining cross-domain gt sampling and class-balance re-sampling with Bi3D++. This is because only a few rare-class instances can be obtained by random sampling, the model will learn more knowledge from past source domain instances using cross-domain gt sampling or learn from duplicated frames with rare-class using class-balance re-sampling. As a result, the model will interfere with domain discrepancies or overfit to the few frames with rare-class instances. However, Bi3D++ can sample diverse frames with more rare-class instances, thus avoiding these situations.

4) Combining With Different Detection Architecture: In Section IV-B, we conduct experiments on two popular detectors. Here we show more results on the pillar-based method (i.e., PointPillar [1]) which is based on a different detection architecture. As shown in Table VIII, we report the results on Waymo $\rightarrow$ KITTI scenario and set the annotation budget to 1%. It can be seen that Bi3D++ still outperforms other ADA methods and significantly reduces the performance gap between the source-only and the oracle (*e.g.*, +20.19% / +22.11 compared to the source-only result). The results further show that our proposed method can be combined with different types of detection architectures.

TABLE VIII
PERFORMANCE ON POINTPILLAR. THE EXPERIMENTS ARE CONDUCTED ON WAYMO $\rightarrow$ KITTI SCENARIO AND WE REPORT AP_{BEV} / AP_{3D}.

Method	Car	Pedestrian	Cyclist
Source Only	53.39 / 10.45	46.93 / 42.96	33.26 / 27.64
TQS	74.59 / 60.13	48.65 / 44.87	49.36 / 46.27
CLUE	75.80 / 60.58	48.73 / 43.71	50.74 / 46.65
Bi3D	77.48 / 64.21	52.16 / 48.64	50.96 / 47.35
Bi3D++	**78.48** / **64.71**	**55.52** / **51.11**	**53.45** / **49.75**
Oracle	80.32 / 70.48	51.34 / 46.68	60.74 / 53.59

TABLE IX
GENERALIZATION OF DIFFERENT METHODS. THE EXPERIMENTS ARE CONDUCTED ON WAYMO $\rightarrow$ KITTI SCENARIO USING PV-RCNN AS THE BASELINE DETECTOR AND WE REPORT THE RESULTS ON BOTH SOURCE AND TARGET DOMAINS. **BOLD**: BEST. UNDERLINE: SECOND BEST.

Method	Source	Target	Mean
Source Only	**49.18** / **42.94**	53.93 / 36.21	51.55 / 39.57
ReDB	41.02 / 30.96	66.11 / 64.45	53.56 / 47.70
SSDA3D	42.67 / 32.15	63.07 / 56.57	52.87 / 44.36
Bi3D	45.64 / 36.86	69.97 / 63.79	57.80 / 50.32
Bi3D++	45.91 / 37.88	**72.11** / **67.55**	**59.01** / **52.20**

5) Analysis of Generalization Ability: DA methods often encounter large performance drops on the source domain when adapting the model to the target domain, since the model will forget some source domain knowledge during the adaptation process. In this part, we show that Bi3D++ has a stronger generalization ability that can achieve better performance on both source and target domains due to the use of selected bi-domain samples. Table IX shows the performance on both source and target domains. It can be observed that our Bi3D++ can outperform other DA methods on both domains at the same time. This is because 1) Bi3D++ selects diverse target data, which largely improves the performance on the target domain and 2) the selected source data are also utilized to train the model during the adaptation process, which prevents the model from forgetting the source knowledge.

6) Combining Hybrid Source Sampling Strategy With Existing ADA Works: We conducted additional experiments combining our designed source sampling strategy with previous ADA methods to further validate the effectiveness of our approach. As shown in Table X, it can be observed that our hybrid active source strategy can effectively combine with previous ADA methods to bring additional performance improvements, which shows the flexibility and effectiveness of the hybrid active source sampling strategy. In addition, when previous ADA methods are combined with our source domain sampling, Bi3D++ can still achieve better results, which further demonstrates that our hybrid target

TABLE X
RESULTS OF COMBINING HYBRID ACTIVE SOURCE SAMPLING STRATEGY WITH PREVIOUS ADA METHODS. S.S. REPRESENTS THE HYBRID ACTIVE SOURCE SAMPLING STRATEGY. WE CONDUCT EXPERIMENTS USING SECOND ON WAYMO ← KITTI.

Method	Car	Pedestrian	Cyclist	Mean
Source Only	51.48 / 19.78	40.80 / 31.26	47.63 / 35.45	46.64 / 28.83
TQS	80.45 / 64.75	50.98 / 47.97	51.43 / 47.05	60.95 / 53.59
TQS + S.S.	81.45 / 65.35	51.26 / 48.37	51.85 / 47.46	61.52 / 53.72
CLUE	82.23 / 65.85	51.46 / 47.56	53.71 / 50.57	62.46 / 54.66
CLUE + S.S.	82.42 / 65.97	51.86 / 47.93	**53.77** / 50.85	62.68 / 54.91
Bi3D++	**85.62** / **71.07**	**54.98** / **50.41**	53.68 / **51.47**	**64.75** / **57.65**

TABLE XI
RESULTS ON NUSCENES → WAYMO USING PV-RCNN AS DETECTOR

Method	Vehicle	Pedestrian	Cyclist
Source Only	39.18 / 22.88	3.27 / 2.38	1.24 / 1.07
ReDB	52.21 / 33.35	6.77 / 5.35	4.29 / 3.88
SSDA3D	60.96 / 49.70	35.51 / 31.73	33.48 / 32.75
Bi3D	61.28 / 50.36	41.84 / 37.56	35.25 / 32.73
Bi3D++	**63.43** / **53.96**	**46.85** / **43.66**	**36.48** / **33.94**
Oracle	70.36 / 59.05	57.85 / 51.51	51.12 / 49.07

TABLE XII
SENSITIVITY TO THE HYPERPARAMETERS ON WAYMO → KITTI USING PV-RCNN AS DETECTOR

α	AP_{BEV} / AP_{3D}	β	AP_{BEV} / AP_{3D}
0.9	70.66 / 66.31	0.9	70.34 / 66.25
0.99	**71.04** / **66.62**	0.99	**71.04** / **66.62**
0.999	70.98 / 66.55	0.999	70.55 / 66.52

domain sampling strategy can effectively sample valuable target domain data, thereby achieving superior performance.

7) Results on Label-Insufficient Domain to Label-Rich Domain: To further evaluate the ability of Bi3D++, we conduct experiments on a label-insufficient source domain to a label-rich target domain. Since KITTI only has annotations in the front view (FOV) which may cause interference with detection in other areas of the target domain, we choose nuScenes as the source domain and Waymo as the target domain. We only sample 0.33% instead of 1% when Waymo serves as the target domain to reduce the annotation costs. As shown in Table XI, our Bi3D++ also achieves the best performance on the label-insufficient domain to label-rich domain setting, which also shows that our Bi3D++ can handle different cross-domain settings.

8) Sensitivity to Hyperparameters: In this part, we ablate the impact of hyperparameters α and β, which are associated with the update of the memory bank B_t and B_r. As shown in Table XII, we set α and β to 0.999, 0.99, and 0.9, respectively. Note that when ablating on α, we set $\beta = 0.99$ and when ablating on β, we set $\alpha = 0.99$ as our default setting. It can be observed that the fluctuation of the performance is 0.38% / 0.31% and 0.70% / 0.37% when changing α and β, respectively. The results show that our method is robust to these hyperparameters.

9) Qualitative Results: We further provide the qualitative results of using different sampling strategies to show the effectiveness of Bi3D++. It can be seen that while the source sampling strategy can alleviate domain differences to some extent, there are still many instances where objects are either unrecognized or inaccurately recognized. Through the target sampling strategy, the model can be effectively transferred to the target domain.

V. CONCLUSION

In this paper, we propose a hybrid bi-domain active learning approach, namely Bi3D++, to tackle cross-domain 3D object detection tasks. To design an effective approach, we analyze the challenges in domain adaptive 3D object detection tasks and design a two-stage active learning method. In the active source sampling stage, Bi3D++ combines scene-level (i.e., domainness) and instance-level similarity to narrow the distribution gap between the source and target domains. In the active target sampling stage, Bi3D++ is composed of a rare-class similarity criterion to sample sufficient frames with rare-class instances and intra-frame and inter-frame diversity criteria to sample frames including diverse representations. Extensive experiments on different cross-domain settings demonstrate that Bi3D++ can outperform existing UDA, SSDA, and ADA methods and can be combined with these works flexibly.

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